



Semester 2 Examinations 2018/2019

Exam Code(s)	3BMS2, 3BPT1, 3BS9, 3BPT1, 3FM2, 4BME1, 4BMS2, 4BS2
Exam(s)	4 th Arts, 3 rd & 4 th Science
Module Code(s)	MP307
Module(s)	Modelling II, Mathematical Modelling
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor Y. Fu
Internal Examiner(s)	Dr M. Meere Professor M. Tuite

Instructions: Marks are awarded for question 1 in Section A **and** the TWO best questions in Section B.
All questions carry the same weight.

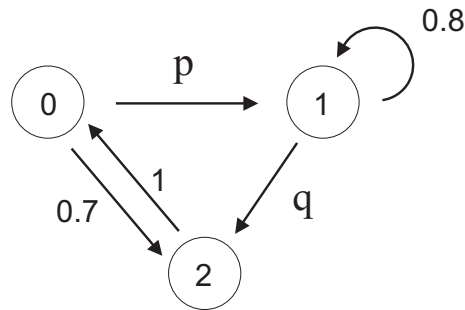
Duration	2 hours
No. of Pages	6
Discipline(s)	Applied Mathematics
Course Co-ordinator(s)	Prof. M. Tuite

Requirements:

Release in Exam Venue	Yes	☒	No	☐
Handout	None			
Statistical/ Log Tables	Yes			
Graph Paper	None			
Log Graph Paper	None			
Other Materials	Calculator			

Section A - Answer all parts (a)-(d). (Each part is worth **5 Marks**.)

1. (a) Describe what is meant by a finite ergodic Markov process with transition probability matrix P . Define the equilibrium probabilities Π_i for a finite ergodic system and show that they obey the matrix equation $\Pi = \Pi P$. Describe a sufficient condition for P which guarantees ergodicity.
- (b) Consider the following Markov Process with transition probabilities shown.



- (i) Find the transition matrix P and the probabilities p and q .
- (ii) Is the Markov process ergodic? Explain your answer.
- (iii) Does the Markov process have unique equilibrium probabilities? Explain your answer. If the Markov process has unique equilibrium probabilities calculate them.

PTO

- (c) Consider an ecological system with a prey population of size $x(t)$ and a predator population of size $y(t)$ which satisfy the differential equations

$$\begin{aligned}\frac{dx}{dt} &= x(-by + a), \\ \frac{dy}{dt} &= y(bx - c),\end{aligned}$$

for constants $a, b, c > 0$.

- (i) Give an interpretation for the various terms in these equations.
 - (ii) Locate all equilibrium points.
- (d) Consider a linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= a_1x_1 + a_2x_2 + b_1u, \\ \frac{dx_2}{dt} &= a_3x_1 + a_4x_2 + b_2u \\ y &= c_1x_1 + c_2x_2\end{aligned}$$

where x_1 and x_2 represent the state, $u \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output, and $a_1, a_2, a_3, a_4, b_1, b_2, c_1, c_2$ are real constants.

- (i) Rewrite the linear system in matrix form.
- (ii) Describe what is meant by *reachability* for this system.

Section B - Answer any two questions.

2. Assume that the size of a queue is given by a nearest-neighbour Markov model and that it has a Poisson arrival pattern with parameter α_k and a Poisson servicing pattern with parameter β_k , where k is the queue size.

- (a) Let the maximum size of the queue be r .
(i) Explain why $\alpha_r = \beta_0 = 0$.
(ii) Assuming that the Markov Process is ergodic, show that the equilibrium probabilities are given by

$$\pi_k = \rho_k \rho_{k-1} \cdots \rho_1 \pi_0$$

for $0 < k \leq r$ where $\rho_k = \alpha_{k-1}/\beta_k$.

8 Marks

- (b) Let the queue in (a) have

$$\alpha_k = \alpha, \quad \beta_k = k\beta,$$

where $\alpha > 0$ and $\beta > 0$.

Show that the equilibrium probabilities π_k , $k = 0, 1, 2, \dots, r$, are given by

$$\pi_k = \frac{\rho^k}{k!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \cdots + \frac{\rho^r}{r!}}$$

where $\rho = \alpha/\beta$.

6 Marks

- (c) Let the queue in (b) correspond to a phone exchange with 4 operators. If 24 people call the exchange per hour and each phone call is serviced in 5 minutes, on average. What is the probability that the exchange is at capacity, i.e. each operator is engaged?

6 Marks

3. Two competing species with population sizes P_1, P_2 evolve according to the following differential equations

$$\begin{aligned}\frac{dP_1}{dt} &= 4P_1(6 - P_1 - P_2), \\ \frac{dP_2}{dt} &= P_2(2 - P_1 - P_2).\end{aligned}$$

- (a) Give an interpretation for the various terms in these equations.

4 Marks

- (b) Find the equilibrium points for P_1, P_2 .

4 Marks

- (c) Show that for any initial non-zero populations $P_1(t) \rightarrow 6$ and $P_2(t) \rightarrow 0$, as $t \rightarrow \infty$. What is the interpretation of this result?

12 Marks

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_1,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0, p_1, p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the concentration of the drug in compartment 2.

- (a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.

4 Marks

- (b) Investigate for what values of p_0, p_1 and p_2 this system is observable.

6 Marks

- (c) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $r = 2$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -1$ and $\lambda_2 = -1$.

10 Marks